Matchmaking

Queue request fields, polling states, tilt handling, cancellation, and navigation.

Overview

Matchmaking places a registered player in a server-side queue using a selected time control. The client joins once, then polls the server every 1.5 seconds until it receives a matched game ID or an error.

Request fields

Source: data/remote/api/MatchmakingService.kt

Field	Type	JSON name	Default	Meaning
timeControlMinutes	Int	time_control_minutes	required	Initial minutes per player.
incrementSeconds	Int	increment_seconds	required	Increment after a move.
rated	Boolean	rated	true	Rated or casual queue.
ignoreTilt	Boolean	ignore_tilt	false	Retry after a tilt warning.

```
POST /api/matchmaking/join Body: { time_control_minutes, increment_seconds, rated, ignore_tilt }
```

Response fields

Field	Type	JSON name	Meaning
state	String?	state	Expected values include searching, matched, idle.
gameId	String?	game_id	Online game ID when matched.
error	String?	error	Server error description.
tiltWarning	TiltWarning?	tilt_warning	Optional anti-tilt warning.

Tilt warning fields

Field	Type	Meaning
tilted	Boolean	Whether the warning is active.
reasons	List<String>	Reasons supplied by the server.
suggestions	List<TiltSuggestion>	Suggested actions.
TiltSuggestion.type	String	Suggestion category.
TiltSuggestion.message	String	Human-readable suggestion.

Client state machine

```
startMatch() -> isSearching = true -> POST /api/matchmaking/join state=matched -> matched(gameId) ->
navigate online_game/{gameId} state=searching -> poll GET /api/matchmaking/status every 1500 ms
state=error -> fail(message) Polling searching -> continue matched -> stop polling, expose matchedGameId
idle -> expire search exception -> show reconnecting message and continue
```

Endpoints

HTTP	Path	Method	Use
POST	api/matchmaking/join	joinQueue	Enter queue.
POST	api/matchmaking/cancel	cancelQueue	Leave queue.
GET	api/matchmaking/status	getStatus	Poll queue status.

Files and responsibilities

File	Responsibility
presentation/game/screens/MatchmakingScreen.kt	Renders searching, error, tilt-warning, cancel, and matched states.
presentation/game/viewmodel/MatchmakingViewModel.kt	Starts queue request, polls, handles cancellation, retry, and errors.
data/remote/api/MatchmakingService.kt	Retrofit request/response models and endpoints.
presentation/navigation/AppNavigation.kt	Validates account/time control and opens matchmaking route.
data/remote/AuthInterceptor.kt	Refreshes access token before authenticated REST requests.

Access rules

The navigation layer blocks guests and unauthenticated users before opening matchmaking. A valid registered access token is required. Authenticated Retrofit services share the access-token refresh and 401 retry path.